KRAS G12C

Covalent Tractability Map v1.0

AXARA Covalent 30 — SDA Photoaffinity Probe Site Analysis

Target:	KRAS G12C
UniProt:	P01116 | Gene: KRAS
Primary Structure:	6OIM | 1.65 A | RCSB Quality: REVIEW
Scoring Mode:	POCKET
Top Probe Site:	CYS12 | Score: 10.05 | 3P Confidence: 3.0
Known Covalent Site:	C12
Validation:	PASS | Rank #1
ChEMBL:	CHEMBL2095731 | 0 compounds | Min IC50: None nM
PDB Covalent:	487 deposited covalent structures
UniProt Sites:	0 active | 22 binding

Third-party data sources: fpocket v4.0 (Le Guilloux 2009) · UniProt Swiss-Prot (UniProt Consortium 2023) · ChEMBL (Mendez 2019) · RCSB PDB Validation (wwPDB 2011) · Shannon conservation (Shannon 1948) · BioPython Shrake-Rupley SASA (Shrake & Rupley 1973)

Covalent Probe Site Rankings

All nucleophilic residues scored by AXARA Engine v5.0. Scored: 32 | Excluded: 14 | Mode: POCKET. Third-party score (0-10) combines RCSB quality, UniProt annotation, conservation, and ChEMBL/PDB covalent evidence.

Rank	Res	Pos	SASA	Score	3P	Cons	UniProt	Evidence
1	**CYS**	12	15.8	**10.05**	3.0	0.0	BND	VALIDATED — known co
2	**LYS**	169	158.1	**8.34**	4.0	1.0		NOVEL PREDICTION
3	**LYS**	104	67.4	**7.63**	4.0	1.0		NOVEL PREDICTION
4	**TYR**	64	131.5	**7.08**	4.0	1.0		NOVEL PREDICTION
5	**LYS**	101	64.0	**6.51**	4.0	1.0		NOVEL PREDICTION
6	**LYS**	88	70.2	**6.0**	4.0	1.0		NOVEL PREDICTION
7	**LYS**	167	60.6	**5.71**	4.0	1.0		NOVEL PREDICTION
8	**HIS**	95	36.8	**5.17**	4.0	1.0		NOVEL PREDICTION
9	**LYS**	165	55.3	**5.13**	4.0	1.0		NOVEL PREDICTION
10	**SER**	136	20.4	**5.09**	4.0	1.0		NOVEL PREDICTION
11	**LYS**	128	84.8	**4.9**	4.0	1.0		NOVEL PREDICTION
12	**SER**	39	19.3	**4.78**	4.0	1.0		NOVEL PREDICTION
13	**HIS**	166	26.9	**4.73**	4.0	1.0		NOVEL PREDICTION
14	**LYS**	117	6.6	**4.72**	4.67	0.833	BND	LITERATURE-ADJACENT
15	**TYR**	40	36.1	**4.35**	4.0	1.0		NOVEL PREDICTION
16	**TYR**	32	48.7	**4.34**	5.0	1.0	BND	LITERATURE-ADJACENT
17	**TYR**	71	14.5	**4.22**	3.67	0.833		NOVEL PREDICTION
18	**LYS**	147	30.0	**3.98**	3.67	0.833		NOVEL PREDICTION
19	**HIS**	27	89.0	**3.88**	4.0	1.0		NOVEL PREDICTION
20	**SER**	65	6.4	**3.81**	4.0	1.0		NOVEL PREDICTION

Third-Party Validation Summary

Source	Data	Value	Significance
UniProt Swiss-Prot	P01116	0 active sites · 22 binding sites	Independent biological annotation
RCSB PDB Quality	6OIM	1.65 A · Clash 1.46 · Rama 0.0%	wwPDB certified: REVIEW
ChEMBL	CHEMBL2095731	0 compounds · 0 covalent · Min IC50: None nM	Published medicinal chemistry landscape
RCSB Covalent Search	P01116	487 covalent structures deposited	Direct experimental evidence
Shannon Conservation	UniProt sequence	169 highly conserved positions	Evolutionary validation (Shannon 1948)

fpocket v4.0	6OIM	Pocket detection + druggability · Mode: POCKET	Le Guilloux et al. 2009 (>2000 citations)